How to Define Expected Quantum Polynomial-Time Zero Knowledge Simulation

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Zero-knowledge proofs

An interactive protocol (P, V) is a zero-knowledge proof for L if:

- Completeness: $\forall x \in L, \Pr[\langle P(x), V(x) \rangle \to 1] = 1.$
- Soundness: $\forall x \notin L \wedge \tilde{P}, \Pr\left[\left\langle \tilde{P}(x), V(x) \right\rangle \to 1\right] \leq \epsilon.$
- Zero-knowledge: $\forall \tilde{V}, \exists S_{\tilde{V}}$ such that $\text{View}_{\tilde{V}}\left(P(x), \tilde{V}(x)\right) \approx S_{\tilde{V}}(x).$

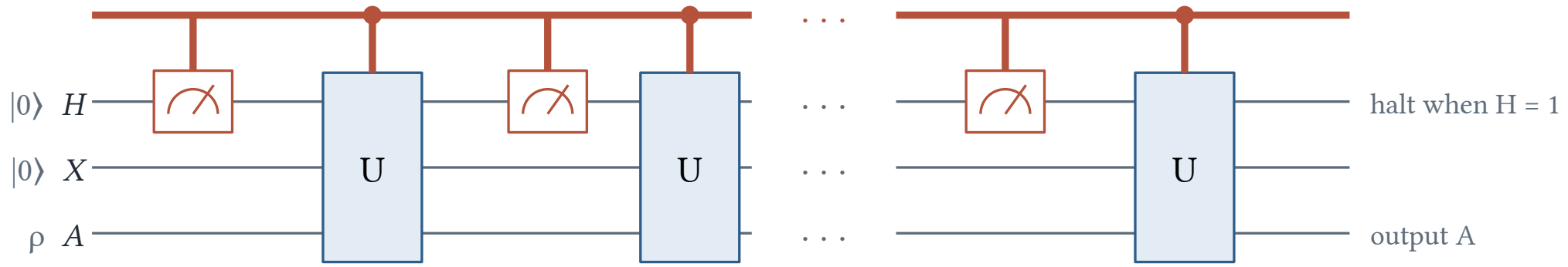
Defining zero-knowledge simulators

For constant-round, black-box zero-knowledge with negligible errors…

Simulator	View of classical verifier	View of quantum verifier
Strict polynomial time	Only for BPP [BL02]	Only for BQP [CCLY21, CCLL26]
Expected polynomial time	For GNI: GMW For NP: Feige–Shamir, Goldreich–Kahan	**Not clear how to define it**

Goal: find the "right" definition of expected quantum polynomial time (EQPT) in this setting.

A first attempt: EQPT_m computation



EQPT_m (measured-runtime EQPT):
expected number of steps is polynomial

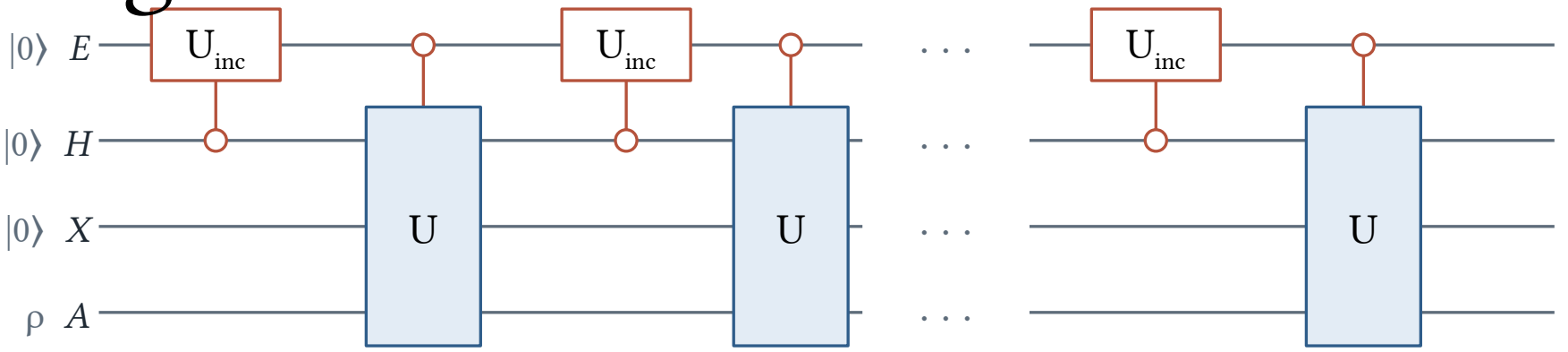
- [CCLY21, CCLL26]: Constant-round black-box ZK with EQPT_m simulation only exists for languages in BQP.
- The measurement collapses the state of the computation, preventing useful interference between branches of different runtime.

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Storing the runtime



"Canonical" dilation of an $EQPT_m$ computation M ([LMS22])

- The clock register *E* counts the steps taken after the computation halts.
- The computation is exactly M when tracing out the ancilla register *E*.

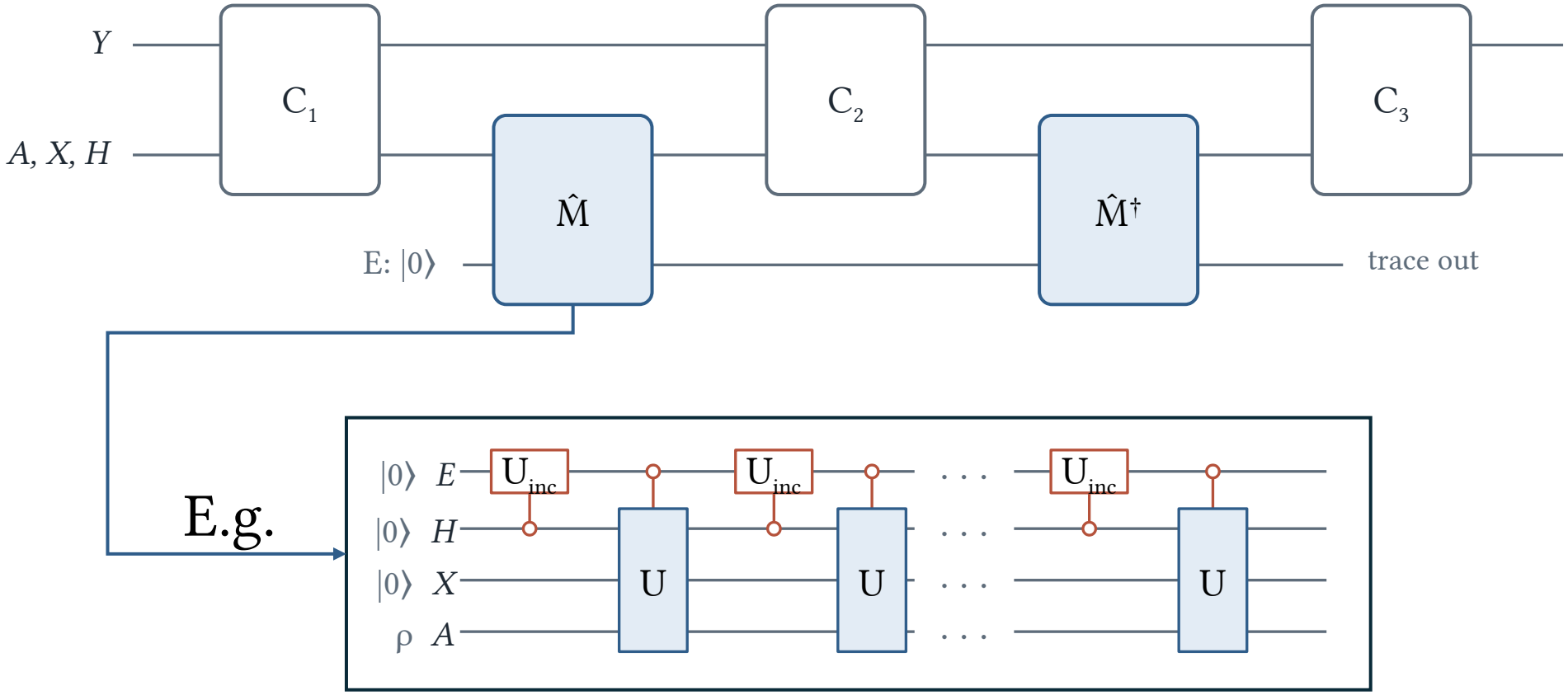
PRO

The circuit records the runtime and has an inverse!

CON

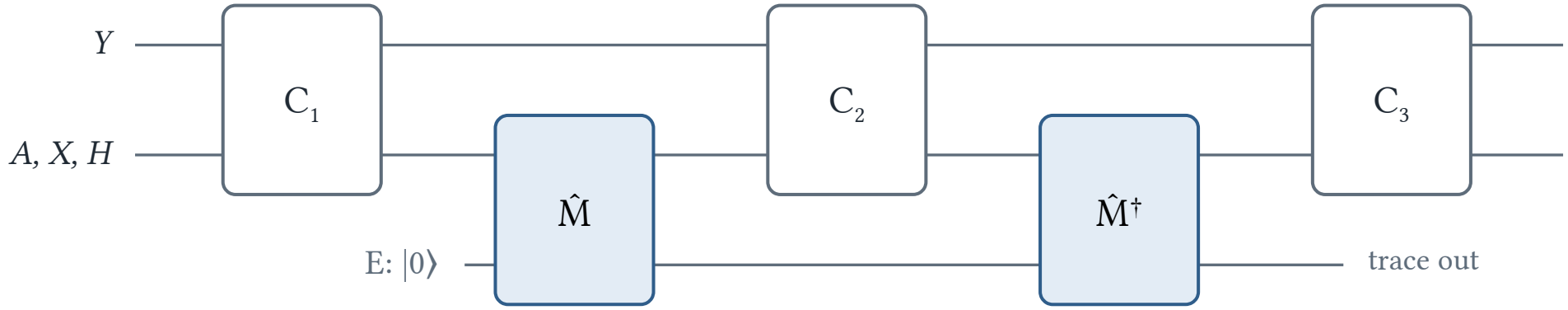
The circuit is exponentially large…

EQPT_c computation [LMS22]



C_i: poly-size circuits. $\hat{M}$: a unitary dilation of an EQPT_m procedure M (Lombardi–Ma–Spooner 2022)

EQPT_c computation [LMS22]



1: existing protocols have EQPT_c simulators

[LMS22] showed this for the GMW, Feige–Shamir and Goldreich–Kahan protocols.

2: EQPT_c captures only efficient computations

[LMS22] shows this for the circuit with canonical dilation, and claims that it holds for all dilations because of unitary invariance.

Our contributions

1 The flaw

EQPT_c isn't well-defined

The output depends on which dilation $\hat{M}$ is used, and there can be an EQPT_c circuit that **decides any NP problem.**

2 Our fix

Restrict the dilations

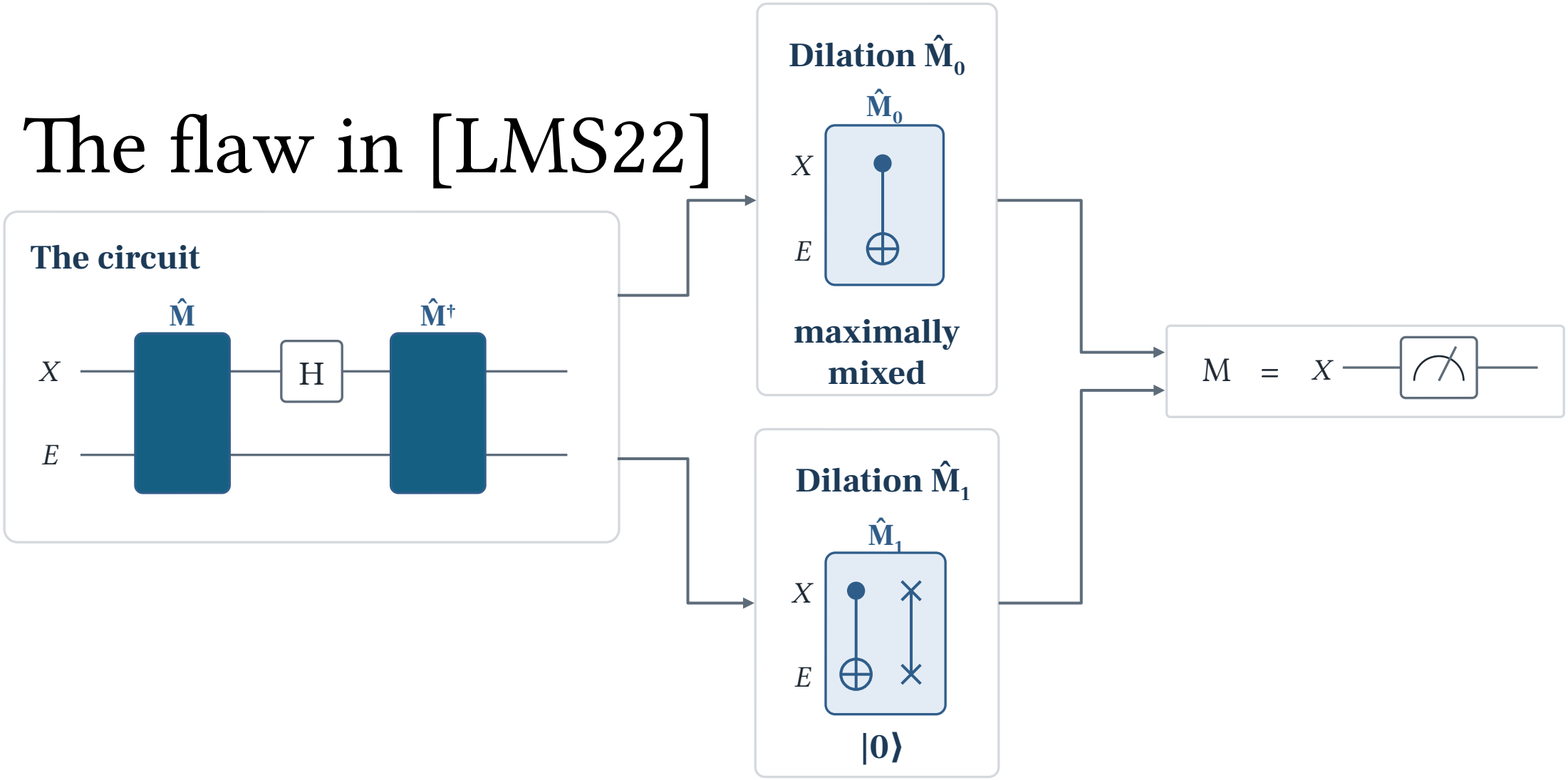
We require correctness for only **a class of dilations**. GMW, Feige-Shamir and Goldreich–Kahan stay Zero-knowledge.

3 New result

ZK proofs for QMA

Constant-round (4-message) ZK proof for QMA against malicious verifiers with negligible errors.

The flaw in [LMS22]



- The output of an EQPT_c circuit is **NOT invariant under the choice of** $\hat{M}$.
- **Even worse**, there are EQPT_c circuits that solves any NP problem using carefully designed unitary dilation. (See example in our paper)

Our fix

ε-ZK

Definition: A protocol is ε-ZK if the simulator runs in time poly(1/ε) and has error up to ε.

$EQPT_c$-ZK
with canonical dilation

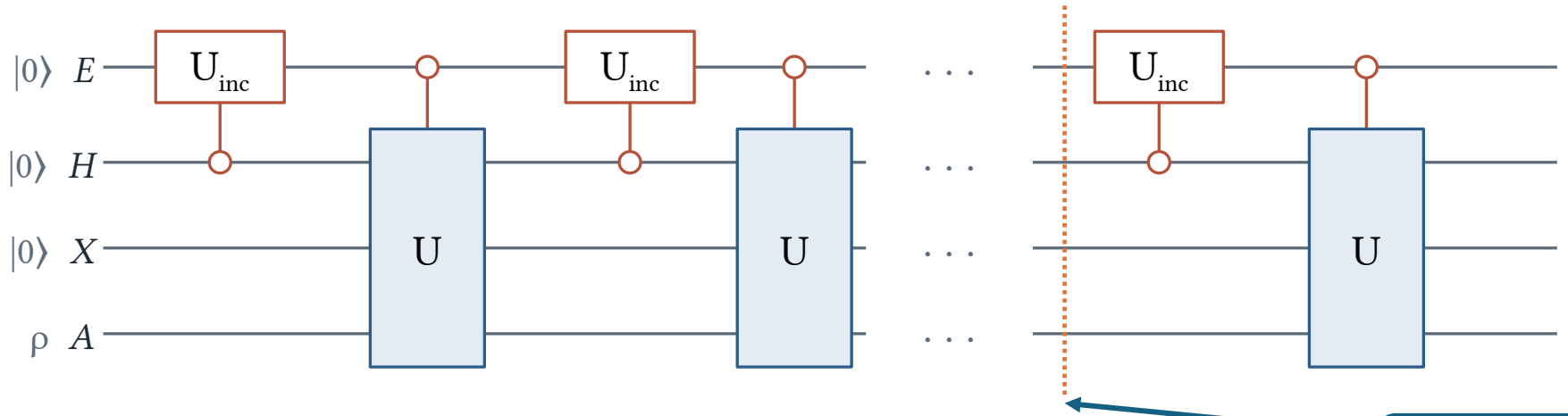
Implies ε-ZK, and relatively easy to prove correctness of the simulator.

Seems overly restrictive.

$EQPT_c$-ZK
with a family of *efficiently approximable* dilations

The simulator correctness extends to any dilation in the family!

Efficiently approximable dilations



Definition (Informal). A unitary dilation is **efficiently approximable** if it can be approximated to arbitrary γ closeness by a poly(1/γ)-size circuit.

Under this definition, one can show that the simulator correctness extends to any unitary dilation within this family.

Protocol	Language
GMW graph non-isomorphism	GNI
Feige–Shamir 4-message argument	NP
Goldreich–Kahan 5-message proof	NP
Our 4-message proof (fully quantum)	QMA

Idea:

Apply the Goldreich-Kahan template initialized with quantum commitments, which saves one message of interaction.

The rewinding technique of the simulator has the same idea as in [LMS22], with lemmas adapted in the quantum setting instead of post-quantum.

Takeaways

https://ia.cr/2026/1500

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Simulator runs in…	Classical settings	Quantum settings
Strict polynomial time	Only for BPP [BL02]	Only for BQP [CCLY21, CCLL26]
Expected polynomial time	For GNI: GMW For NP: Feige–Shamir, Goldreich–Kahan	EQPT_m (BQP); **EQPT_c** ✅

Alternatively, ε-ZK:

Simulator runs in…	Classical settings	Quantum settings
Strict polynomial time	[DNS98]	[CCY21, LMS22]